Complex Numbers

ANSWER KEY

Section 1: Complex Arithmetic & Polar Form

1 What is the polar (trigonometric) form of the complex number $z = -2 + 2i\sqrt{3}$?

a. $4(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$ ✓

b. $4(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

c. $2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

d. $4(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

$r = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4$. Angle θ is in Quadrant II: $\theta = \pi - \arctan(\sqrt{3}) = \frac{2\pi}{3}$.

2 What is the product of $z_1 = 3(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$ and $z_2 = 2(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$?

a. $6(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$ ✓

b. $5(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

c. $6(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

d. $6(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8})$

Multiply magnitudes and add angles: $r = 3 \times 2 = 6$, and $\theta = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$.

3 Simplify the quotient $\frac{8(\cos 80^\circ + i\sin 80^\circ)}{2(\cos 20^\circ + i\sin 20^\circ)}$.

a. $4(\cos 60^\circ + i\sin 60^\circ)$ ✓

b. $4(\cos 100^\circ + i\sin 100^\circ)$

c. $4(\cos 4^\circ + i\sin 4^\circ)$

d. $6(\cos 60^\circ + i\sin 60^\circ)$

Divide magnitudes and subtract angles: $\frac{8}{2} = 4$, and $80^\circ - 20^\circ = 60^\circ$.

Section 2: De Moivre's Theorem & Powers

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4 Using De Moivre's Theorem, evaluate $(1+i)^8$.

a. 16 ✓

b. $16i$

c. 32

d. -16

$1+i=\sqrt{2}(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4})$. Raising to power 8:
$(\sqrt{2})^8[\cos(8\cdot\frac{\pi}{4})+i\sin(8\cdot\frac{\pi}{4})]=16(\cos 2\pi+i\sin 2\pi)=16(1+0)=16$.

5 What is $(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6})^{12}$?

a. 1 ✓

b. -1

c. i

d. $-i$

By De Moivre's Theorem: $\cos(12\cdot\frac{\pi}{6})+i\sin(12\cdot\frac{\pi}{6})=\cos(2\pi)+i\sin(2\pi)=1$.

6 What is the real part of the complex number $(2-2i)^4$?

a. -64 ✓

b. 64

c. 0

d. -32

$2-2i=\sqrt{8}e^{-i\pi/4}$. Raising to 4th power: $(\sqrt{8})^4e^{-i\pi}=64(-1)=-64$.

Section 3: Roots of Complex Numbers & Roots of Unity

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7 How many distinct complex solutions exist for the equation $z^5 = 1$?

a. 1

b. 3

c. 5 ✓

d. Infinitely many

By the Fundamental Theorem of Algebra, any polynomial of degree n has exactly n complex roots. Here $n = 5$.

8 Which of the following is a primitive cube root of unity?

a. $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2} + i\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2} - i\frac{1}{2}$

d. $1 + i\sqrt{3}$

Roots of $z^3 = 1$ are $e^{i2k\pi/3}$. For $k = 1$: $\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

9 What is the sum of all the n-th roots of unity for any integer $n \geq 2$?

a. 0 ✓

b. 1

c. n

d. -1

The roots form a symmetric geometric progression whose sum is $\frac{1-\omega^n}{1-\omega} = \frac{1-1}{1-\omega} = 0$.

Section 4: Complex Polynomials & Geometry

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10 If $2 + 3i$ is a root of the quadratic polynomial $x^2 + bx + c = 0$ with real coefficients, what is the value of $b + c$?

a. 9 ✓

b. 11

c. 13

d. 17

By the Complex Conjugate Root Theorem, the other root is $2 - 3i$. Sum of roots $= 4 = -b \Rightarrow b = -4$. Product of roots $= 2^2 + 3^2 = 13 = c$. Thus $b + c = -4 + 13 = 9$.

11 In the complex plane, what geometric figure is described by the equation $|z - (3 + 4i)| = 5$?

a. **A circle centered at** $(3, 4)$ **with radius** 5 ✓

b. An ellipse with foci at $(3, 4)$

c. A line passing through $(3, 4)$

d. A parabola with vertex $(3, 4)$

$|z - z_0| = r$ represents the locus of all points whose distance from $z_0 = 3 + 4i$ is constant (5), which is a circle of radius 5.

12 What is the minimum value of $|z - 6|$ given that $|z| = 2$?

a. 2

b. 4 ✓

c. 6

d. 8

By the reverse triangle inequality: $|z - 6| \geq ||6| - |z|| = |6 - 2| = 4$.

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